GYANMANJRI INNOVATIVE UNIVERSITY

College Helpdesk
Rule-Based Chatbot

Python and Streamlit Assignment Report

Student name	_____
Enrollment number	_____
Subject	_____
College	Gyanmanjri Innovative University (GMIU)
GitHub repository URL	_____
Live chatbot URL	_____

Submitted as an academic project demonstrating a transparent, rule-based conversational interface.

1. Introduction

A college helpdesk chatbot can guide students toward information about common administrative and academic topics. This project implements a small, explainable chatbot in Python. It identifies keywords in a student's message and returns a predefined response using conditional statements. A Streamlit interface displays the conversation in a browser.

The chatbot is an educational demonstration, not an official GMIU information system. It does not have access to student records and avoids making up current fees, dates, phone numbers, or policies.

2. Objectives

- Build a working rule-based chatbot in Python.
- Use keyword matching and explicit if, elif, and else conditions.
- Provide fixed answers for ten college helpdesk topic groups and common social messages.
- Create a clear, responsive chat interface with examples and a clear-chat option.
- Direct students to official college sources when a fact may change or is not verified.

3. Tools and Technologies

Tool	Use in this project
Python	Implements keyword matching and response selection.
Streamlit	Creates the interactive browser-based chat interface.
Regular expressions (re)	Normalizes text and matches complete keywords.
ReportLab	Generates this PDF assignment report.

The chatbot's answer logic runs locally in the application process. It does not call an external AI API or machine-learning model.

4. Rule-Based Chatbot Logic

The program first normalizes the message by converting it to lowercase, removing punctuation, and collapsing repeated spaces. It then checks the normalized message against keyword groups. Each branch is an if or elif condition; the first matching branch returns its fixed response. The final else branch handles questions that do not match any known pattern. Empty text has its own first condition.

The matching order puts helpdesk topics before greetings. For example, “Hi, what are the fees?” should reach the fee rule rather than stop at a greeting rule.

Algorithm

```
START
Read the user's message
Normalize text (lowercase; remove punctuation)
IF message is empty: ask for a question
ELIF admission keywords match: show admissions guidance
ELIF fee keywords match: direct to official fee notice/accounts office
ELIF another supported topic matches: show its fixed response
ELIF greeting / thanks / goodbye matches: show social response
ELSE: show the unknown-question response
Display response in the chat
END
```

Simplified decision flow

User message	Normalize	Keyword branch	Fixed answer
Typed or clicked example	Lowercase + punctuation removed	First matching if/elif rule	Displayed in chat

5. Important Parts of the Source Code

The actual project contains the complete matching rules and interface. This short excerpt demonstrates the main conditional structure:

```
if not text:
return "Please type a question, and I'll try to help."
elif has_keyword(text, ("admission", "admissions", "application", "enroll")):
return RESPONSES["admissions"]
elif has_keyword(text, ("fee", "fees", "tuition", "payment")):
return RESPONSES["fees"]
else:
return (
"I'm not sure about that question yet. Please rephrase it or contact "
"the relevant GMIU department through an official college channel."
)
```

The helper function has_keyword checks for a complete word or phrase, reducing accidental matches such as finding “hi” inside a longer unrelated word. Streamlit stores conversation messages in session state so they remain visible across interface reruns.

6. Sample Conversations

The following bot messages match the predefined response strings in app.py. The sample includes more than the required seven conversations.

User question	Chatbot response
Hello	Hello! What would you like to know about GMIU?
How do I apply for admission?	Admissions dates and eligibility can change by program and academic year. Please confirm the latest notice with GMIU's official admissions portal or admissions office.
What are the current fees?	Fee amounts and payment deadlines vary by program and academic year. For the current fee structure, check the official GMIU fee notice or contact the accounts office. I don't have verified current figures.
Where can I see the timetable?	Timetables can differ by department, semester, and section. Check your department's official timetable notice or student portal for the current schedule.
When are the examinations?	Exam dates and instructions may change. Please check the latest examination notice on GMIU's official student portal or contact the examination office.
How do I check my result?	Results are published through the official student or examination portal. Please verify there; I can't access personal result records.
Can I borrow books from the library?	For library hours, catalogue access, borrowing limits, and renewals, check the library notice or contact GMIU's library staff.
What scholarship information is available?	Scholarship eligibility, deadlines, and available schemes vary. Check the latest official scholarship notice and confirm details with the scholarship or financial aid office.
How do I apply for a hostel?	Hostel availability, fees, rules, and application dates can change. Confirm current details directly with the hostel or administration office.

User question	Chatbot response
When are holidays?	Holiday schedules are set by the university and may change. Refer to the current academic calendar or an official college notice.
How can I contact the college?	For verified phone numbers and email addresses, use the contact information listed on GMIU's official website or contact your department office. I don't want to guess an official number.
Thank you	You're welcome! Let me know if you have another college helpdesk question.

7. Testing

The rule function was checked against representative prompts, including blank and unrecognized input. PASS means the returned answer matched the intended response category and avoided fabricating official information.

Test input	Expected response	Result
How do I apply for admission?	Admissions guidance; verify current dates/eligibility.	PASS
What are the current fees?	Official fee notice/accounts office; no unverified amount.	PASS
Where can I see the timetable?	Check the department notice or student portal.	PASS
When are the examinations?	Check the official examination notice/office.	PASS
How do I check my result?	Use the official portal; no personal result access.	PASS
Can I borrow books from the library?	Check library notice/staff for current rules.	PASS
What scholarship information is available?	Check official scholarship notice/office.	PASS
How do I apply for a hostel?	Confirm current details with hostel/administration office.	PASS
When are holidays?	Check the current academic calendar or official notice.	PASS
How can I contact the college?	Use verified official website/department contact details.	PASS
(blank input)	Ask the user to type a question.	PASS
What is the moon made of?	Unknown-question response; rephrase or contact the relevant office.	PASS

8. Advantages

- Simple and explainable: each response is linked to a visible keyword rule.
- Fast and lightweight: no model, database, API key, or paid AI service is required.
- Predictable: the same normalized question follows the same matching order.
- Safer handling of changing college details: uncertain facts are referred to official sources.

9. Limitations

- The chatbot recognizes only keywords and phrases included in its rules.
- It cannot understand context, paraphrases it has not been taught, or follow-up intent like a conversational AI model.
- Its fixed answers do not contain live schedules, fee amounts, admissions status, or personal student results.

- A message containing multiple topics follows the first matching topic according to the rule order.

10. Future Improvements

- Add more tested keywords and alternate phrasings based on common student questions.
- Use a reviewed, versioned FAQ dataset with an administrator update process.
- Add multilingual responses and accessible interface improvements.
- Add optional feedback controls and report which questions were not recognized, without collecting private student information.

11. Conclusion

This project demonstrates how a basic rule-based chatbot can answer common college helpdesk questions through keyword matching and if/elif/else conditions. The Streamlit interface makes the program easy to try in a browser. Since college information can change, the chatbot avoids unsupported official claims and directs students to GMIU's official notices and offices for verification.

12. Repository and Live App

GitHub repository URL	_____
Live chatbot URL	_____

Fill in these fields only after creating the repository and successfully deploying the application. Do not submit example or guessed URLs as real links.